



Storage Devices

HDD, SSD, USB, CD/DVD, Memory Cards and Cloud Storage.

5.1 What is Storage?

5.2 Units of Storage

5.3 Storage Devices



What We Cover Today



Introduction to Computers



Parts of a Computer (Hardware)



Software



Input & Output Devices

05

Storage Devices

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Practice & Weekly Quiz

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Operating Systems

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Windows Desktop

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Mouse & Keyboard Skills

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File Management

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Windows Settings

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Installing & Uninstalling Software

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Touch Typing Fundamentals

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File Sharing, Cables & USB

15

Computer Networks

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System Management & Troubleshooting

▲ **TODAY** ✓ **COMPLETED**

Recap: Unit 4



The IPO Cycle

Every task follows INPUT → PROCESS → OUTPUT. Keyboard → CPU → Monitor.



Input Devices

Send data TO the computer. Keyboard, Mouse, Scanner, Webcam, Microphone.



Output Devices

Give results FROM the computer. Monitor, Printer, Speaker, Projector.



Both Input & Output

Touchscreen and Headset do both — they send data in and give results out.

? Quick check: is a Touchscreen an input device, an output device, or both?

Today's Objectives



Define Storage

Explain the difference between Primary and Secondary storage.



Learn Storage Units

Convert between Bit, Byte, KB, MB, GB, TB and PB.



Identify Storage Devices

Recognise HDD, SSD, USB, Memory Cards, Optical Discs and Cloud Storage.



Get Hands-On

Check your laptop's storage, a file size, and organise files into a folder.

 We'll close with 3 hands-on tasks and a quick recap — keep these four goals in mind.



UNIT 05

Storage Devices

HDD, SSD, USB, CD/DVD, Memory Cards and Cloud Storage.



What is Storage?

Storage is where the computer keeps data — permanently or temporarily. Without storage, all your files, programs and the OS would be lost the moment you switch off.



Primary Storage (Main Memory)

TEMPORARY

SPEED

Very fast — CPU can access it directly

DURATION

Temporary — data is LOST when computer is off

LOCATION

Inside the computer (on the motherboard)

CAPACITY

Small — typically 4 GB to 32 GB

EXAMPLE

RAM (Random Access Memory)

*RAM stores data the computer is currently using.
When you close a program, that data disappears from RAM.*



Secondary Storage (Auxiliary)

PERMANENT

SPEED

Slower than RAM but much more capacity

DURATION

Permanent — data is KEPT when computer is off

LOCATION

Inside or connected to the computer

CAPACITY

Large — 128 GB to several TB

EXAMPLES

HDD, SSD, USB Flash Drive, CD/DVD

*Secondary storage keeps your files safe permanently.
Your photos, documents and OS all live here.*

Units of Storage

Every piece of data is measured in units — from a single BIT to a PETABYTE.

BIT

Smallest unit — a 0 or a 1

BYTE

= 8 Bits

KB

Kilobyte = 1,024 Bytes

MB

Megabyte = 1,024 KB

GB

Gigabyte = 1,024 MB

TB

Terabyte = 1,024 GB

PB

Petabyte = 1,024 TB



Think of it like this:

8 Bits make 1 Byte. After that, every unit is exactly 1,024 times bigger than the one before it — Byte → KB → MB → GB → TB → PB. That's because computers count in powers of 2, and $1,024 = 2^{10}$.



What Fits in Each Unit?

1 KB

One short text message or SMS

1 MB

One song (MP3) or about 500 pages of text

1 GB

One standard-quality movie or 250 songs

1 TB

250,000 songs or about 1,000 full movies

1 PB

All data of a large company's servers

Your laptop

Usually 256 GB, 512 GB or 1 TB (SSD/HDD)

Secondary Storage Devices

These devices store your data permanently — even when the computer is switched off.



HDD

Hard Disk Drive

📦 500 GB – 4 TB ⚡ Slower (spinning disk)

Uses magnetic spinning disks to store data. Very common in desktop computers.

e.g. Desktop PCs, external backup drives



SSD

Solid State Drive

📦 128 GB – 2 TB ⚡ Very fast (no moving parts)

Uses flash memory chips. Faster, lighter and quieter than HDD. Found in modern laptops.

e.g. Laptops, gaming PCs, phones



USB

USB Flash Drive / Pen Drive

📦 4 GB – 256 GB ⚡ Moderate speed

Small portable device that plugs into a USB port. Easy to carry and use.

e.g. Transferring files between computers



Memory Card

SD / MicroSD Card

📦 2 GB – 1 TB ⚡ Moderate speed

Tiny card used in cameras, phones and tablets to store photos, videos and apps.

e.g. Smartphones, cameras, drones



Optical Disc

CD / DVD / Blu-ray

📦 700 MB – 50 GB ⚡ Slow

Uses a laser to read and write data. Stores music, movies or software.

e.g. Movies, music albums, software install



Cloud

Cloud Storage

📦 5 GB – Unlimited ⚡ Depends on internet

Stores files on remote internet servers. Access from any device anywhere in the world.

e.g. Google Drive, OneDrive, Dropbox

Task 1 — Check Your Laptop's Storage

Exercise 1 of 3

 **Goal:** Find out how much storage your laptop has and how much is used.

1

Step 1

Click Start → type This PC → open. Look at Local Disk (C:). You will see a bar showing used space (blue) and free space. Hover the mouse over the bar to read the numbers.

2

Step 2

Write down three numbers: (1) Total capacity in GB. (2) Used space in GB. (3) Free space in GB. This is your laptop's SSD or HDD — the main secondary storage device!

3

Step 3

☒ Check: Tell your teacher — (1) Is your storage SSD or HDD? (Hint: modern laptops have SSD.) (2) Is your drive mostly full or mostly empty? (3) Is 256 GB more or less than 1 TB?

Task 2 — Check a File Size

Exercise 2 of 3



Goal: Save a Notepad file and find out its size in KB.

1

Step 1

Open Notepad. Type 5 sentences about anything you learned this week. Then save: File → Save → choose Desktop → type the filename: myfile.txt → click Save.

2

Step 2

Close Notepad. On the Desktop, find myfile.txt. Right-click on it → click Properties. Look for Size — you will see the file size in KB (kilobytes). How small is it?

3

Step 3

☒ Check: Your text file is very tiny (a few KB). Your teacher will show you a photo file — it will be in MB (megabytes). This shows why we need different units of storage!

Task 3 — Create a Folder

Exercise 3 of 3

 **Goal:** Create a folder on the Desktop and save your file inside it.

1

Step 1

On the Desktop, right-click on an empty area. Hover over New → click Folder. A new folder appears. Type the name: My Files → press Enter. Your folder is created!

2

Step 2

Find the myfile.txt you saved earlier. Left-click on it and drag it into the My Files folder. Double-click the folder to open it and check that your file is inside.

3

Step 3

☒ Check: Is myfile.txt inside the My Files folder? You just organised files on secondary storage! Your laptop stores all files and folders exactly like this on the HDD or SSD.



Key Takeaways



What is Storage?

Storage holds data permanently or temporarily. Primary (RAM) is fast but temporary. Secondary (HDD/SSD/USB) is permanent — your files stay safe when the computer is off.



Units of Storage

8 Bits = 1 Byte. Then $\times 1024$ each time: Byte $\rightarrow$ KB $\rightarrow$ MB $\rightarrow$ GB $\rightarrow$ TB $\rightarrow$ PB. A song is ~ 1 MB. A movie is ~ 1 GB. Your laptop usually has 256 GB – 1 TB of storage.



Physical Storage Devices

HDD: magnetic disk, large capacity, affordable. SSD: flash memory, very fast, found in modern laptops. USB Flash Drive: portable, plugs in anywhere. Memory Card: cameras and phones.



Optical & Cloud Storage

CD/DVD/Blu-ray: stores music, movies and software using a laser. Cloud Storage: saves files on the internet — access from anywhere. Examples: Google Drive, OneDrive, Dropbox.

Next up — Unit 6: Practice & Weekly Quiz



What You Learned Today

Tasks 1 to 3 complete — everything from this lesson in one place.

STORAGE DEVICES — REFERENCE



Primary vs Secondary

PRIMARY (RAM)

- Fast access speed
- TEMPORARY — lost when off
- Small capacity (4–32 GB)
- Example: RAM

SECONDARY

- Slower than RAM
- PERMANENT — kept when off
- Large capacity (256 GB–TB)
- HDD, SSD, USB, Cloud

Units of Storage

BIT (0/1) → BYTE (=8 Bits) → KB → MB → GB → TB
📦 (each ×1,024)



Storage Devices



HDD

500 GB–4 TB

Magnetic disk, desktops



USB Flash

4–256 GB

Portable, plug & use



CD / DVD

700 MB–50 GB

Music, movies, software



SSD

128 GB–2 TB

Fast, modern laptops



Memory Card

2 GB–1 TB

Phones & cameras



Cloud

5 GB–∞

Google Drive, OneDrive

Primary storage is temporary (RAM). Secondary storage is permanent (HDD, SSD, USB, Cloud).